

LINGERING TAILS

GENOME WIDE ASSOCIATION STUDY

Study Design

INPUT

N=434



GREEN ASH



HYBRIDS



WHITE ASH



PHENOTYPING



RESISTANT



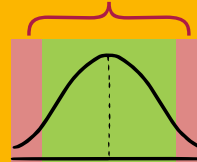
SUSCEPTIBLE

Analysis

- Output
 - Association tests
 - Odds ratios or effect sizes
 - Manhattan plot
 - Visualizes SNPs across the genomes that
- Identify what genes and gene regions the SNPs of interest originate from
- Determine function and pathways of those genes
 - How are they relevant in the trees reaction and defense to the pest?

Tails

These are the tails:
the most resistant and
most susceptible



Most trees have
susceptibility that fall
within this range

We are interested in the most resistant tails. By conducting a GWAS on them, we hope to identify relevant SNPs, then genes, that are important in EAB resistance.

Bioinformatics

1. Input data
 - a. WGS (15x) of 434 green ash, white ash, and hybrids
 - b. Sampled during the major die-offs
2. Quality control
 - a. Check sequencing quality
3. Align to reference
 - a. Zane has a white and green genome that has been annotated*
 - b. Use VaryCool
4. Quality control for missingness, pop structure, relatedness, linkage, etc.
5. Run a million different GWAS with different covariates, groupings (white vs green, both, etc.)

**Do QTL analysis to
verify findings**